

COVID-19 in India

Exploratory data analysis and forecasting of national case counts, 2020-01-22 to 2023-03-09

1. Summary

- India recorded **44,690,738** confirmed cases and **530,779** deaths in the period, a reported fatality ratio of 1.19%.
- Three distinct waves were detected. The second wave was the largest: the 7-day average peaked at **391,232** cases/day on 2021-05-08, about 4.2x the first wave's peak.
- The third wave (Omicron) was the fastest. The 7-day average grew x21.6 in 14 days (ending 2022-01-12), yet its crude deaths-to-cases ratio (0.54%) was far lower than earlier waves (1.43% and 1.35%).
- In a 14-day rolling backtest, ARIMA(2,1,2) and Damped Holt had the lowest sMAPE (15.8%) against 22.5% for a naive baseline. Gradient boosting had the lowest MAE (8,781 cases/day).
- No univariate model anticipated the speed of the second wave. Forecasts of this kind are useful for short-term trend extrapolation, not for predicting wave onsets.

2. Data and method

Source. The JHU CSSE COVID-19 time series (cumulative confirmed cases and deaths for India). The dataset has national totals only; recoveries stopped being reported in 2021 and are not used. There are no missing dates.

Cleaning. Daily counts are first differences of the cumulative series, with negative corrections clipped to zero. A 7-day moving average removes the weekday reporting rhythm and is used for waves, growth, and forecasting.

Waves. Peaks of the 7-day average are detected with a minimum prominence of 15% of the global maximum and 45 days minimum separation. Wave boundaries are the lowest points between consecutive peaks.

Forecasting. The target is the 7-day average of daily new cases on a $\log(1+x)$ scale. Four approaches are compared: a naive last-value baseline, damped-trend Holt exponential smoothing, ARIMA(2,1,2), and gradient boosting on the last 14 days applied recursively. Evaluation is a rolling-origin backtest with 45 origins (every 21 days from August 2020), each fitted only on data up to the origin and scored on the next 14 days using MAE and symmetric MAPE (sMAPE).

3. Exploratory analysis

3.1 Growth and milestones

The first 1,000 cases were reached on 2020-03-29, and 100,000 on 2020-05-18. One million came on 2020-07-16, ten million on 2020-12-18, and the total passed 40 million on 2022-01-25.

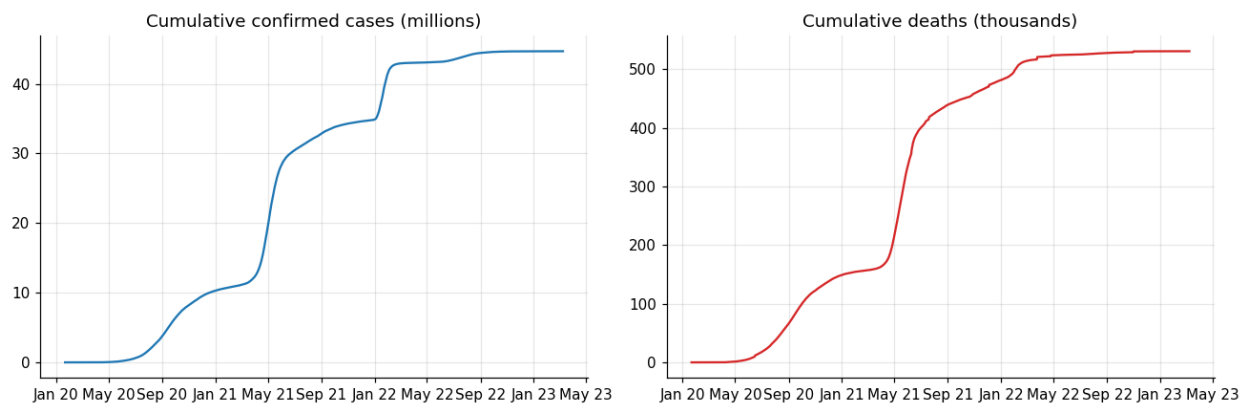


Figure 1. Cumulative confirmed cases and deaths.

3.2 The three waves

Wave	Peak date	Peak 7-day avg	Days to peak	Cases	Deaths	Deaths/cases
Wave 1	2020-09-16	93,198	238	10,880,603	155,448	1.43%
Wave 2	2021-05-08	391,232	86	23,861,900	321,798	1.35%
Wave 3	2022-01-25	311,982	38	9,967,852	53,910	0.54%

Wave 1 built slowly over eight months and was never very sharp. Wave 2 rose in under three months to a peak four times higher. Wave 3 climbed in about five weeks and fell almost as quickly; its long tail runs to the end of the data.

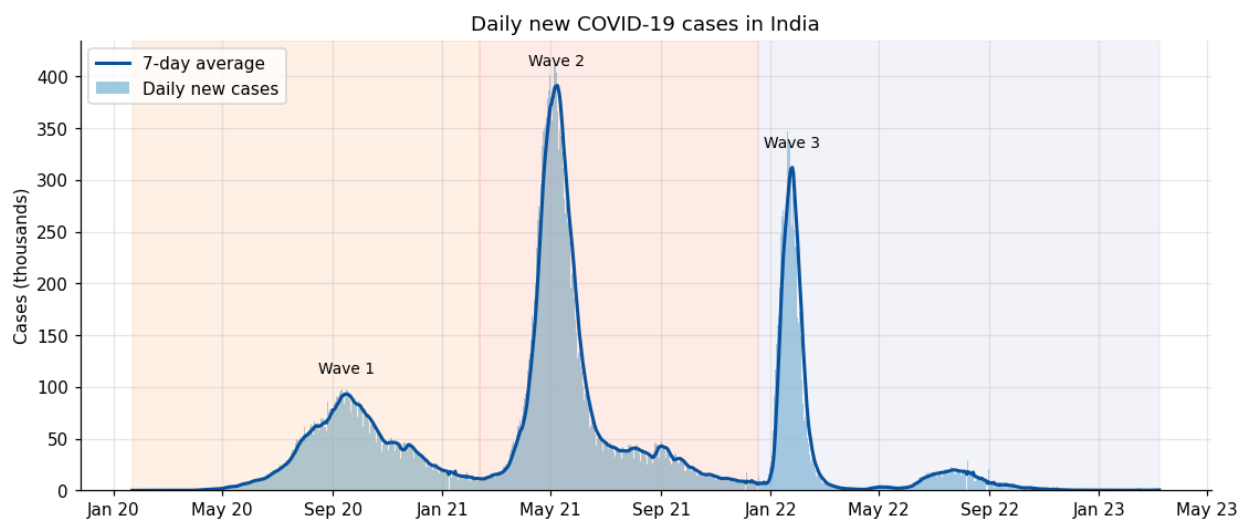


Figure 2. Daily new cases, 7-day average, and the detected waves.

3.3 Deaths and the fatality ratio

The ratio of deaths to confirmed cases fell across the three waves (Figure 3), consistent with a combination of broader testing, vaccination and changing variant severity. It should be read as a reporting ratio, not a true fatality rate. The largest single-day death count in the data (7,374 on 2021-06-10) is a backlog reconciliation, not a real one-day surge.

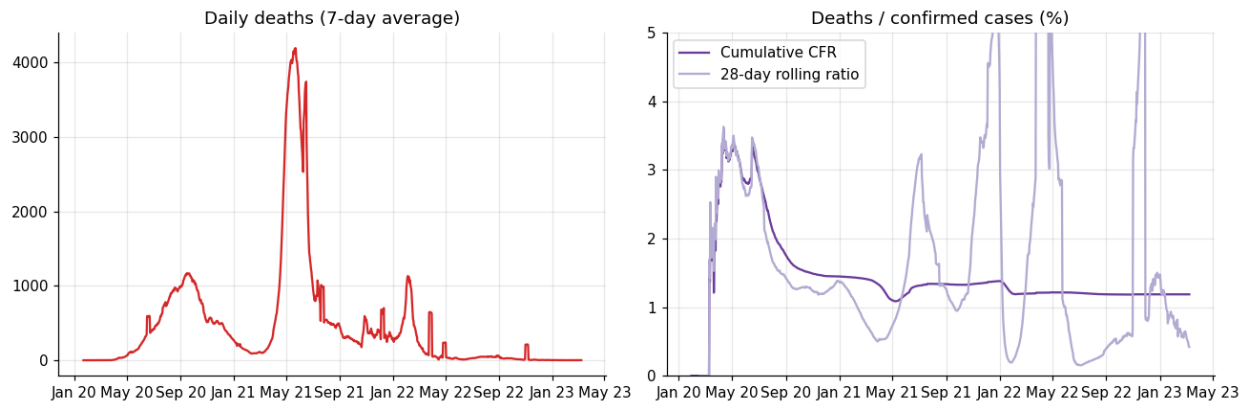


Figure 3. Daily deaths (7-day average) and the deaths-to-cases ratio.

3.4 Growth rate and reporting pattern

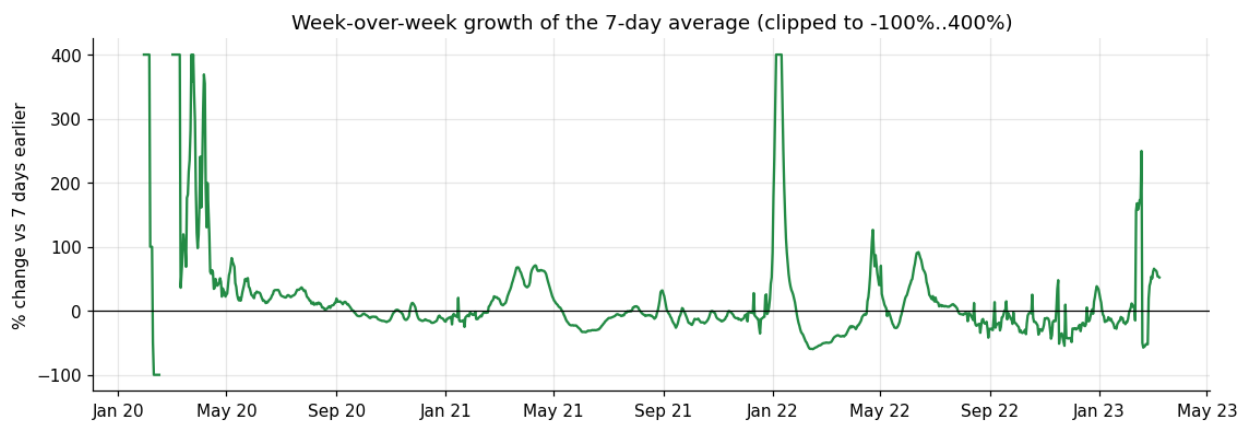


Figure 4. Week-over-week change in the 7-day average (clipped).

Reporting follows a clear weekday rhythm (Figure 5): Monday counts run about 18% below the weekly average, while Wednesday is about 10% above, likely reflecting weekend testing and reporting lags. This is why a 7-day average is used throughout.

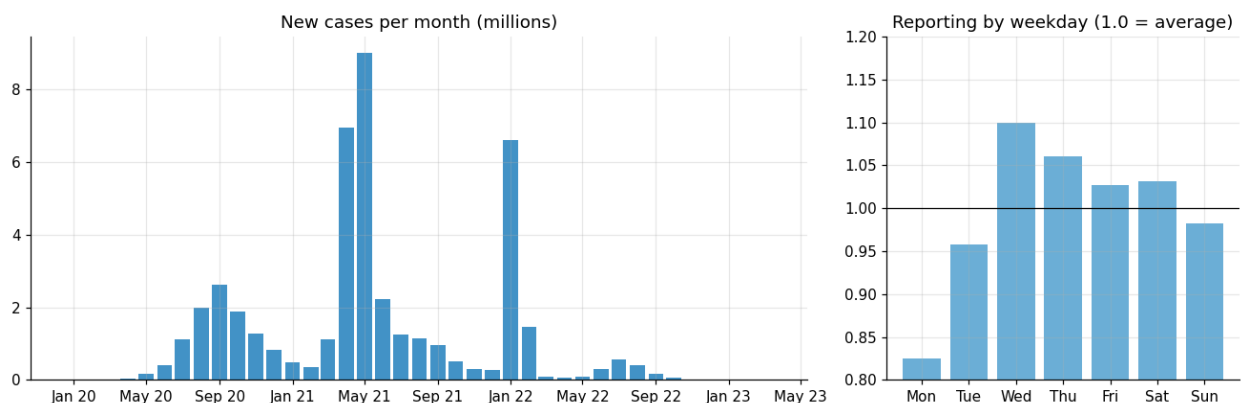


Figure 5. New cases per month, and mean daily reporting ratio by weekday.

4. Forecasting

4.1 Backtest results

Model	Mean MAE	Median MAE	Mean sMAPE	MAE vs naive
ARIMA(2,1,2)	12,815	1,286	15.8%	-1.9%
Damped Holt	13,316	1,313	15.8%	-5.9%
Gradient boosting	8,781	1,494	18.0%	+30.2%
Naive	12,578	2,403	22.5%	+0.0%

The metrics disagree, which is informative. MAE is in cases per day, so it is dominated by the few origins near wave peaks where numbers are large; there gradient boosting does best. sMAPE is scale-free and rewards getting the level right in the quiet periods, where ARIMA and Holt do best. ARIMA's median MAE (1,286) is far below the naive baseline's (2,403), but its mean MAE (12,815) is no better than naive (12,578), because a handful of wave-onset errors dominate the average.

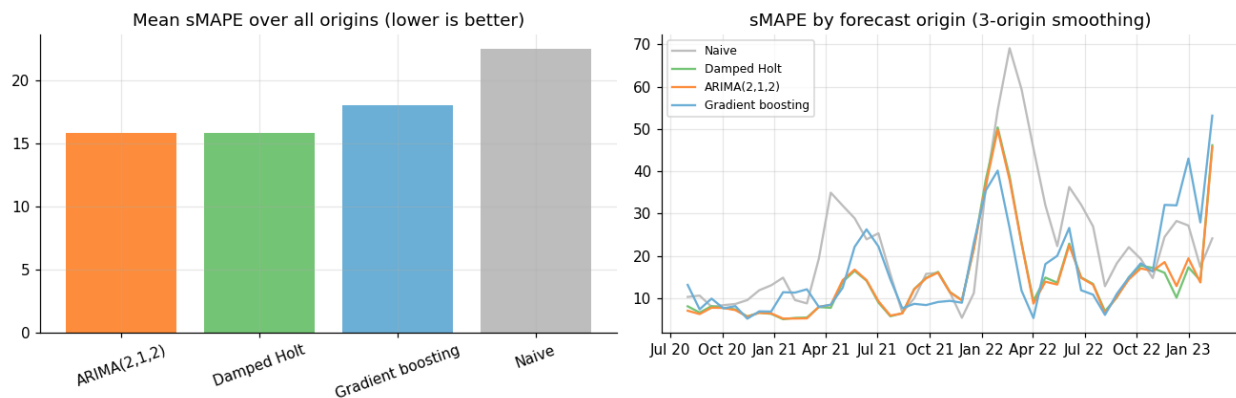


Figure 6. Backtest accuracy by model and by forecast origin.

4.2 Stress test: the start of Wave 2

On 2021-04-05, as the second wave was starting, every model was asked for a 14-day forecast. The actual 7-day average 14 days later was 233,074 cases/day. All models under-forecast: naive 84,313, damped Holt 157,496, ARIMA 154,988, gradient boosting 194,416. The learned momentum in the boosting model helped, but none had the information (mobility, variants, behaviour) that drove the surge.

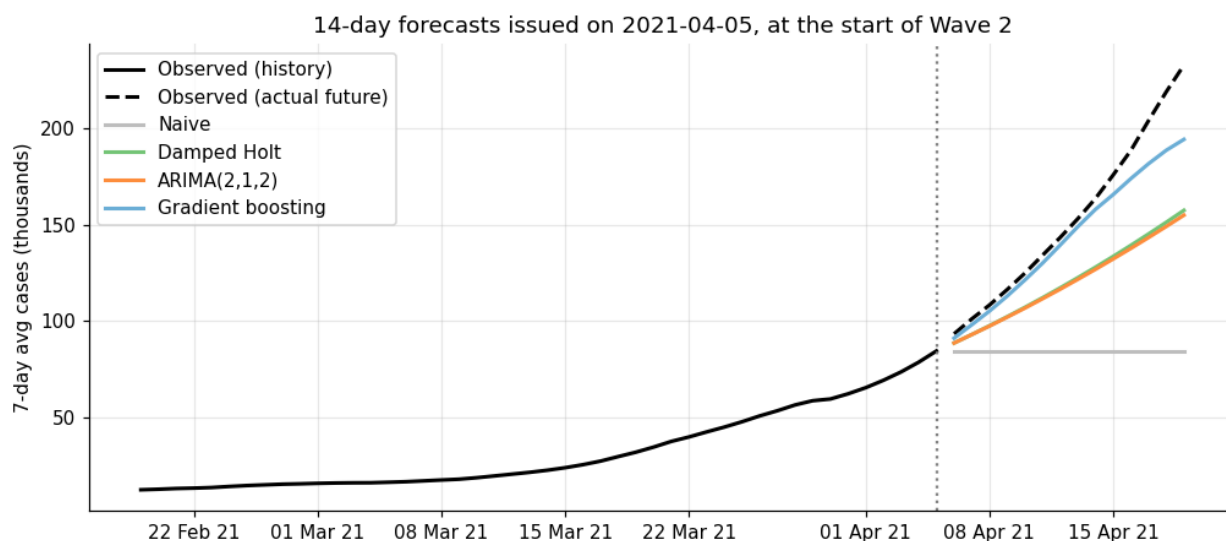


Figure 7. 14-day forecasts issued at the start of Wave 2.

4.3 Forecast beyond the last observation

ARIMA(2,1,2) is used for the final 30-day forecast because it ties for the lowest sMAPE and provides a statistical prediction interval. From a last observed 7-day average of 336 cases/day, the model projects about 740 by 2023-04-08, with a 95% interval of 78 to 6,980. The interval is wide because uncertainty compounds multiplicatively on the log scale. Since the dataset ends in 2023-03-09, this is a demonstration of the method rather than a current prediction.

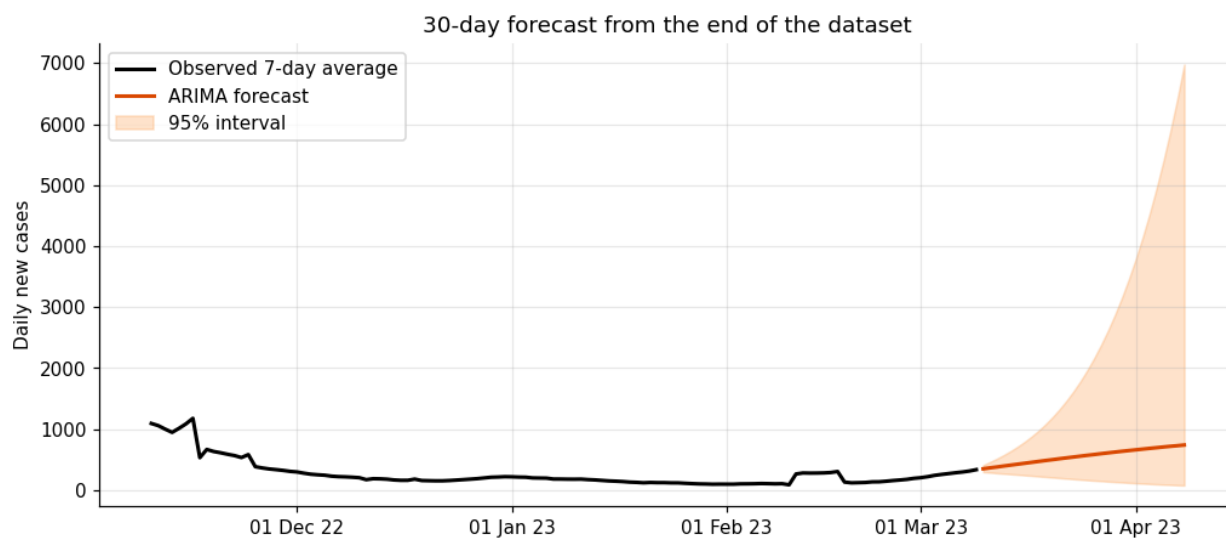


Figure 8. 30-day ARIMA forecast with 95% prediction interval.

5. Limitations

- **Univariate.** Only past case counts are used. Mobility, vaccination, testing and variant data drive wave onsets and are absent.
- **Reported, not true, cases.** Testing intensity and reporting changed substantially; levels are less reliable than trends.
- **National only.** The source has no state-level breakdown, so regional timing differences are invisible.
- **Modest backtest.** 45 origins dominated by three waves; small differences between close models are not statistically strong.